

Gönül Demir

Senior Electronics R&D & Product Engineer

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Hardware, sensing and EMC engineering from concept to certification. Electronics R&D engineer with 15+ years of experience developing optical sensors, electronic security systems, encoders, embedded hardware and industrial products. I turn complex requirements into measurable architectures, prototypes and production-ready solutions, combining circuit design, optics, validation, EMC/EMI and certification.

SELECTED IMPACT

- Developed an optical beam detector with **2.4× range improvement** over local competitors.
- Achieved a **20% range advantage** over the closest global competitor in long-distance optical fire detection.
- Reduced production BOM cost by **30%** through optical and electronic design optimisation.
- Reduced production test time by **15%** and defect rate by **10%** through custom test automation.

PROFESSIONAL EXPERIENCE

Senior Research & Development Engineer 2022–2024

Teknim Electronic · Istanbul

Led hardware development for commercial alarm and access-control products, from architecture and optical design through verification and production support.

- Developed high-performance optical beam and fire-detection solutions.
- Established design baselines for reliability, safety and market competitiveness.

Research & Development Engineer 2017–2022

Fenac Encoders & Sensors · Istanbul

Designed magnetic and optical encoders, wheel encoders, ToF-based safety sensors and industrial interface electronics for civilian and defence applications.

- Designed RS422 differential drivers with 24 V / 200 mA short-circuit protection.
- Resolved EMI/EMC problems in high-noise industrial environments.

Research & Development Engineer 2011–2016

Mavili Electric Electronic · Istanbul

Delivered optical hardware, flame-detection electronics, production test equipment and international certification for industrial fire products.

- Completed optical, electronic and EMC/TSE/LPCB certification work for six beam-detector models.
- Developed four flame-detector models through working prototypes.

OPEN ENGINEERING PROJECT

1–30 MHz, 1.3 kW Solid-State SP2T RF Switch

Openly documented HF switching project developed with Hadrien Theveneau. The series publishes calculations, design

reviews, revisions and staged RF validation.

[Read the engineering notes →](#)

CORE EXPERTISE

Hardware architectureOptoelectronicsOptical sensorsEMC / EMIAnalog designRF & power electronicsEmbedded hardwarePCB designProduct verificationCE / LVD / EMCTSE / LPCBProduction test

TOOLS & METHODS

- Altium Designer, KiCad
- Oscilloscope, VNA and optical test setups
- SPICE and first-principles modelling
- Root-cause analysis and design reviews
- Prototype bring-up and staged validation

EDUCATION

Bachelor of Electrical Engineering

Uludağ University · Bursa, Türkiye

Erasmus — Electronic Engineering

Università degli Studi di Salerno · Italy

LANGUAGES

Turkish — Native

English — Professional working proficiency

WORKING STYLE

Evidence-led, transparent and product-focused. I make assumptions explicit, identify risks early and design verification into the architecture.

Open to thoughtful collaborations, challenging R&D projects and product-focused engineering opportunities in hardware, sensing and EMC.